

RTL INNOVATE: Digital VLSI Hackfest

organised by EVDC Club in collaboration with
Department of ECE and ECE VSI D&T

Project Title

Design of a Vending Machine Controller with multiple products and
dynamic pricing

Team CREATIONISTS

Members:

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Introduction to the Problem Statement:

A Vending Machine is required, which can provide 3 or more products at the same time keeping in mind that the user will input any amount of coins that can be less than, greater than or equal to the price of the product.

User should get the change as remaining balance after purchase.

Objective:

Min. no. of products offered = 3

Functional Behaviour --

Money handling:

- User inserts coins → Balance increases.
- System tracks total balance.

Product Selection:

- If balance $\geq$ price → dispense the product
- Deduct price from the balance
- Return remaining change

Dynamic Pricing:

- Input-controlled prices (iterable)

Invalid Conditions:

- Insufficient balance → no dispense
- Invalid selection → ignore the input

Verilog Code:

```
module creationists(
    input clk, rst,
    input [1:0] select,
    input [7:0] coin, prodA, prodB, prodC,
    input insert_coin,
    output reg [7:0] change,
    output reg dispense
);

    reg [1:0] state; //2 bit variable for state of ven. machine
    reg [7:0] balance,price;//8 bit input for balance and price of the
//product we have selected

    parameter IDLE    = 2'd0;
    parameter CHECK   = 2'd1;
    parameter S_DISPENSE = 2'd2;//parameters for different states of
//machine

    always @(*) begin
        case(select)
            2'd0:  price = prodA;
            2'd1:  price = prodB;
            2'd2:  price = prodC;
            default: price = 8'd0;
        endcase
    end
end
```

```
always @(posedge clk or posedge rst) begin
```

```
    if(rst) begin
```

```
        state  <= IDLE;
```

```
        balance <= 8'd0;
```

```
        dispense <= 1'b0;
```

```
        change  <= 8'd0;
```

```
    end
```

```
    else begin
```

```
        case(state)
```

```
            IDLE: begin //idle state
```

```
                dispense <= 1'b0;
```

```
                if(insert_coin)
```

```
                    balance <= balance + coin;
```

```
                else if (balance > 0)
```

```
                    state <= CHECK;
```

```
            end
```

```
            CHECK: begin //condition for check state
```

```
                if(balance >= price)
```

```
                    state <= S_DISPENSE;
```

```
                else begin
```

```
                    balance <= 8'd0;
```

```
                    state  <= IDLE;
```

```
                end
```

```
            end
```

```
            S_DISPENSE: begin //condition for dispense state
```

```
                dispense <= 1'b1;
```

```
                change  <= balance - price;
```

```
                balance <= 8'd0;
```

```
                state  <= IDLE;
```

```
            end
```

```

        default: state <= IDLE;
    endcase
end
end
endmodule

```

Testbench:

```

`timescale 1ns/1ps
module tb_creationists;

    reg clk=0, rst;
    reg [1:0] select;
    reg [7:0] coin;
    reg insert_coin;
    reg [7:0] p0=10, p1=15, p2=20;
    wire dispense;
    wire [7:0] change;

    creationists uut(.clk(clk), .rst(rst), .select(select), .coin(coin),
    .insert_coin(insert_coin), .prodA(p0), .prodB(p1), .prodC(p2),
    .dispense(dispense), .change(change));

    always #5 clk = ~clk;

    initial begin
        rst = 1; #10 rst = 0;
        select = 2'd1;
        coin=5; insert_coin=1; #10 insert_coin=0;
        coin=10; insert_coin=1; #10 insert_coin=0;
        #20;

        select = 2'd2;
    end
endmodule

```

```

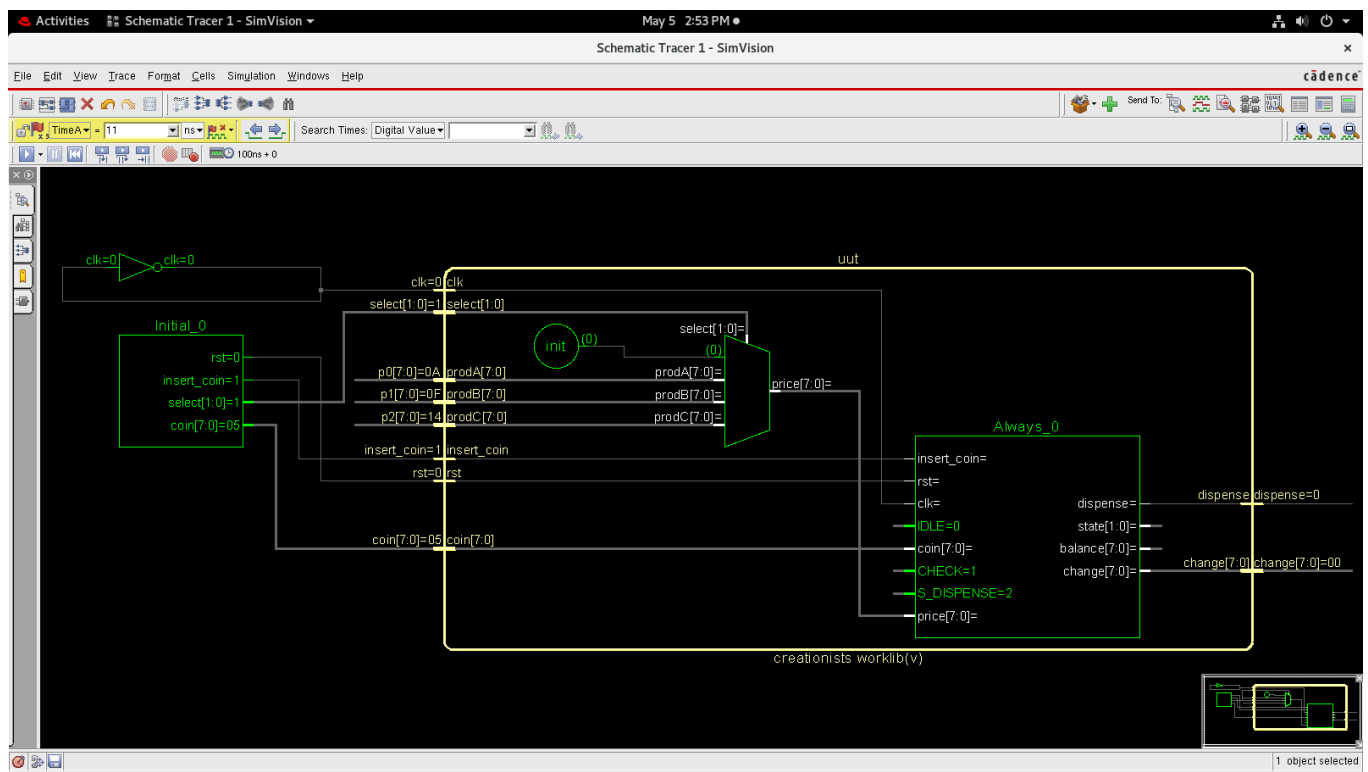
coin=10; insert_coin=1; #10 insert_coin=0;

#40 $finish;
end
initial begin
    $display("\nTime\tSelect\tIns\tCoin\t| Disp\tChange");

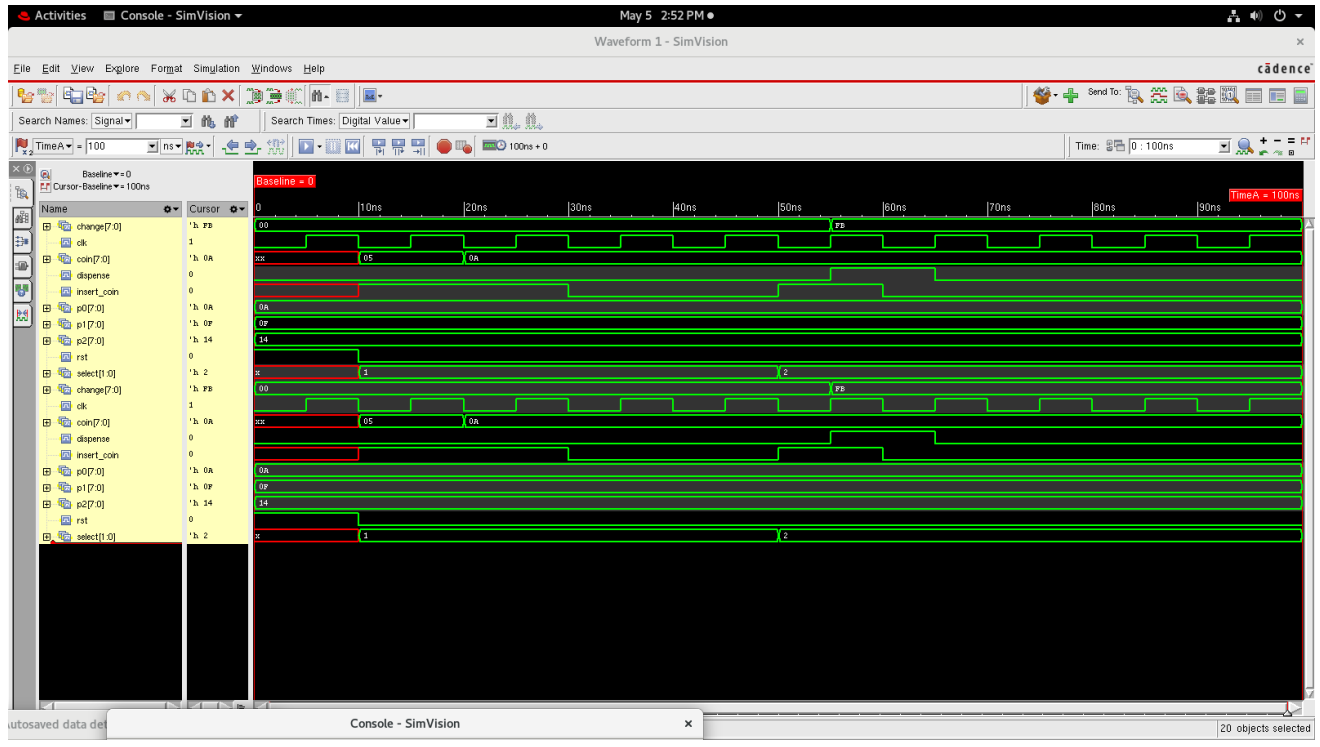
    $monitor("%0t\t%d\t%b\t%d\t| %b\t%d",
        $time, select, insert_coin, coin, dispense, change);
end
endmodule

```

Block Diagram:



Waveform:



Challenges faced:

- FSM Complexity
- Code complexity due to Multiple input-controlled variables

Github Repository Link: [Click to see the Github Repo](#)